

Kyle Main

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Senior Security Data Scientist & Detection Engineer — Elasticsearch-Native ML Threat Detection

12 years of security engineering experience, including 8+ years building Elasticsearch-native detection and analytics platforms and ML-based threat detection content — from an ES/Kibana-based next-gen SIEM at an early-stage startup to a from-scratch DOE/NNSA security data platform with a full UEBA layer on ES transforms. Deep clustering, time-series anomaly detection, and false-positive-reduction ML work, plus hands-on GenAI/LLM production tooling for security automation.

ELASTICSEARCH & ELASTIC SECURITY DEPTH

- DOE/NNSA Security Data Integration: built an entirely new Elasticsearch-based security platform ingesting CrowdStrike, Suricata, and Zeek — including a UEBA detection layer built on ES transforms, custom Kibana dashboards, and native ES data-quality alerting.
- Trend Micro/Cysiv next-gen SIEM: wrote and tuned Elasticsearch Query DSL and native ES detection rules directly against production indexes as core detection content; built 50+ Logstash filters and deployed multiple Elastic Beats variants for log collection into ES.
- CISA CDM at DOE: data ingestion and data-quality engineering across a combined Elasticsearch/Splunk environment supporting a federal SOC.

Full ES lifecycle across three employers: ES API, ES transforms, Query DSL, native detection rules/alerting, Beats collection, and Kibana dashboarding — not just log shipping.

ML & SECURITY DATA SCIENCE

Detection & Anomaly Modeling: Clustering/unsupervised ML for device behavior classification; time-series anomaly detection for entity behaviors (auth volume/geography, process-chain deviations); signature, statistical, behavioral, and ML-based detection rule development with formally tracked false-positive rates and staged rollout

ML Frameworks & Languages: Python, pandas, scikit-learn, NumPy, SciPy, PyTorch, SQL, R

GenAI/LLM for Security: Prompt engineering for detection triage and rule generation; GenAI-driven orchestration of SIEM APIs across 9 platforms; GenAI-powered cross-SIEM rule-conversion tooling for other engineers — hands-on LLM-API-into-production experience

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Treasury SOC (TSSOC), Threat & Research team: own the detection/alerting content (Splunk saved searches) driving live SOC incident response. Previously built DOE/NNSA's Elasticsearch security platform (above) and supported CISA CDM data-quality work at DOE.

Senior Threat Detection Engineer & Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of the data science/detection engineering team building signature, behavioral, statistical, time-series, and ML-based detection content against cloud-scale customer data; incorporated Vedere Labs threat intelligence into detection tuning.

Threat Detection Engineer & Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Early hire building the ES/Kibana-based next-gen SIEM's rules engine and detection content — 2,300+ rules across most of the MITRE ATT&CK matrix; data engineering for 220+ log sources; GCP Dataproc/PySpark exploratory data analysis at scale.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Built DNS-based detection and mitigation for malware infections; analyzed large-scale security log data to surface anomalous behavior.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013), Numerical Data Analysis & Modeling, Applied Physics | B.S. Physics — Ball State University (2011), Minors: Mathematics, Astrophysics

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox